Adrian Eduardo Calderon

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EDUCATION

Florida International University (FIU) | B.S. in Mechanical Engineering, GPA: 3.9 *Expected May 2029*

Miami Dade College | A.A. Degree (Dual Enrollment), GPA: 3.7 *April 2025*

SKILLS

Technical: SolidWorks, Siemens NX, Ansys Mechanical, 3D Printing, Arduino, Python, Angle Grinding, Excel, PowerPoint
Languages: Bilingual (English/Spanish)

EXPERIENCE & LEADERSHIP

Ford Motor Company *Dearborn, MI*
Product Development, Vehicle Engineering Intern *June 2026 – August 2026*

- Oversaw installation of a new component on a future-generation F-600 and performed follow-up verification testing at **-30°C** in a **climatic wind tunnel** with a cross-functional team which validated system performance in extreme cold.
- Earned track certification and independently ran and analyzed a controlled idle **fuel economy test** on a Hybrid Police Interceptor using in-house diagnostic software and a **MyCANIC OBD** interface to benchmark results against previous model-year vehicles.
- Conducted a confidential on-track police pursuit test scenario and evaluated **VBOX GPS data**, delivering a comparative analysis to leadership that demonstrated the distinct performance advantages and tradeoffs of each vehicle.
- Built a **Python** script to automate GPS data transfer from track testing days, improving lap time accuracy by **0.03 seconds** over the prior manual process, and streamlining overall data management into **one** consolidated Excel sheet.

SHPE FIU (Society of Hispanic Professional Engineers) *Miami, FL*
Professional Development E-Board Director *Apr 2026 – Present*

- Lead a **3-person** cabinet to network with professionals, coordinate workshops, and provide members with **1:1** interview prep and resume reviews, expanding opportunities for SHPE members.
- Planned and executed a full-day regional retreat over a 2-month timeline, designing the full event agenda and recruiting **18 industry professionals** for mock interviews and resume reviews; **61 students** from **4 schools** attended.
- Designed and prototyped the idea for ElCamino, a personal internship tracker that auto populates a public job board for members; a computer engineer built the final version, which now holds **350+** listings across **55 members**.

Member Success Coordinator *Sep 2025 – Apr 2026*

- Supported the Professional Development Director in engaging members at events, building relationships across the chapter that led to being elected to the Executive Board the following year.

Panther Motorsports FIU Formula SAE *Miami, FL*
Vehicle Dynamics Engineer *Aug 2025 – Present*

- Performed **FEA** simulations using **Ansys Mechanical** to validate structural integrity of previous brake rotors under maximum braking conditions, revealing a **factor of safety of 8** that indicated the original design was significantly over-engineered.
- Researched braking thermodynamics and steel properties to define a thermal-limited minimum mass, then redesigned the rotors in **SolidWorks**, cutting mass **~30 % front and rear** while re-validating structural integrity in Ansys.
- Conducted component sourcing research to create a custom **GPS** system, including identifying hardware critical to signal transmission, data collection, and device integration.

Levi Strauss & Co. *Miami, FL*
Sales Stylist *Oct 2024 – May 2026 | Aug 2026 – Present*

- Delivered excellent customer service to reach Key Performance Indicators (KPIs), working **30–35** hours a week; maintain an **87%+** satisfaction rate through customer surveys.

TECHNICAL PROJECTS

Arduino-Controlled Robot Arm | Design & Build *Apr 2026 – May 2026*

- Designed and built a functional robot arm controlled via **Arduino**, verifying geometry and constraints (e.g., **180mm** max arm length) in SolidWorks sketches before building to reduce design iterations.
- Diagnosed and resolved hardware failures, including servo mounting issues and cable-housing conflicts, by adapting the design in real time rather than restarting.

RC Airboat | Design & Build *Dec 2025 – Feb 2026*

- Designed and **3D-printed** an RC airboat from scratch; after early iterations failed to float, applied **Archimedes' principle** to redesign hull geometry in **SolidWorks**.
- Increased buoyant force from **5.39N to 6.86N** (exceeding the boat's **6.45N** weight) by growing hull volume from **5.5×10^{-4} m³** to **7.0×10^{-4} m³**, while accounting for ±0.2mm printer tolerances for proper part fitment.

HONORS & AWARDS

- Ford Motor Company | 2026 Intern Better Tomorrow Challenge: Connected Services - Achiever
- Six Sigma White Belt Certificate | Six Sigma Online, 2026
- Florida Bright Futures Scholar (Full-ride, merit-based scholarship)